THE ITERATION $x \mapsto a \bmod x$ HAS LENGTH $O_\varepsilon(a^{23/77+\varepsilon})$

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Abstract. For integers $a > b > 0$ let $b_1 = b$, $b_{i+1} = a \bmod b_i$, and let $P(a,b)$ be the number of positive terms of this sequence. Shallit asked whether $P(a,b) = O((\log a)^2)$. The bound $P(a,b) \ll_\varepsilon a^{1/3+\varepsilon}$ of Erdős and Shallit was improved by Chase and Pandey to $a^{1/3-2/177+\varepsilon}$. We prove $P(a,b) \ll_\varepsilon a^{23/77+\varepsilon}$, where $23/77 = 0.29870\ldots$, by an elementary argument. We first prove $P(a,b) \ll_\varepsilon a^{3/10+\varepsilon}$. The new ingredient there is an exact identity for three consecutive steps of the iteration. It shows that a certain integer combination $D = dv - eu$ of the drops and quotient increments is always positive and is small when the drops are small. A second identity, for six consecutive steps, produces a further positive integer L. It gives the local bound $T \ll a^{1/2}A^{-1/3}$ for the number T of remainders in $[A, 2A)$, with no divisor-function loss. Then D and L turn out to be the third and fourth members of an infinite hierarchy of positive integers: the Hankel determinants of an exact positive moment model of the orbit. To go below $3/10$ we use the arithmetic of L itself. At the critical scale L is a positive integer of bounded size. Seven such integers at two consecutive states form a pattern, and the states with a given pattern lie on a quadric surface in the quotient increments, with at most four states per point. Along the way we show $P(a,b) \ll a^{3/10}\sqrt{\log\log 3a}$, assuming results of Nair–Tenenbaum and de la Bretèche–Tenenbaum on the Erdős–Hooley Δ-function. We also show $P(a,b) \ll a^{1/4}\sqrt{R}$, where $R-1$ is the number of rises in the sequence of quotient increments. Everything is formalised in Lean 4 with Mathlib, without `sorry`. The $\sqrt{\log\log 3a}$ bound is formalised conditionally on the two cited theorems, which are stated in Lean as explicit hypotheses.

1. Introduction

Let $a > b > 0$ be integers. Put $b_1 = b$ and $b_{i+1} = a \bmod b_i$ while $b_i > 0$. The sequence is strictly decreasing until it reaches 0; if $b_{n+1} = 0$ we write $P(a,b) = n$. For example $P(35,22) = 7$, from $22, 13, 9, 8, 3, 2, 1, 0$. The number $P(a,b)$ is the length of the Pierce expansion of b/a [10, 11], since

$$\frac{b}{a} = \frac{1}{q_1} - \frac{1}{q_1 q_2} + \frac{1}{q_1 q_2 q_3} - \cdots, \qquad q_i = \lfloor a/b_i \rfloor.$$

Shallit [9] asked whether

$$P(a,b) = O\big((\log a)^2\big). \tag{1}$$

Shallit [11] proved $P(a,b) \ll a^{1/2}$ by "Archimedean" arguments. Erdős and Shallit [4] combined these with a divisor argument to get $P(a,b) \ll_\varepsilon a^{1/3+\varepsilon}$, and showed that $\max_b P(a,b)/\log a$ does not tend to 0. Chase and Pandey [3] went past the exponent $1/3$, proving $P(a,b) \ll_\varepsilon a^{1/3-2/177+\varepsilon}$ with exponential sum estimates (Bourgain's exponent pair), and showed that $\max_b P(a,b) \gg \log a/\log\log a$ for all large a. Shallit's talk pages [13] report a bound $n^{3/10}$, described as "LLM, unpublished so far". We have not seen that argument. The result below has the same exponent up to ε, and we make no claim about how the two arguments are related.

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Theorem 1.1. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{3/10+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

The proof is elementary. It uses only the division equations $a = q_i b_i + b_{i+1}$ and the divisor bound $\tau(n) \ll_\delta n^\delta$. As usual, the remainders are split into dyadic windows $[A, 2A)$, and we bound the number T of remainders in one window. There are two bounds:

$$T(T-1) \le 4A\,\Delta, \tag{2}$$

$$T^6 A^2 \le 995328\, a^3 \Delta \qquad (T \ge 12), \tag{3}$$

where Δ is the largest value of the divisor function on $[1, 32a^7]$. Bound (2) is the Erdős–Shallit argument. Bound (3) is new. Multiplying (3) by the square of (2) eliminates A and gives $T^{10} \ll a^3\Delta^3$, which is the exponent $3/10$.

The new bound comes from three consecutive steps $x = b_i$, $y = b_{i+1}$, $z = b_{i+2}$, $w = b_{i+3}$. Let d, e, f be the drops, let u, v be the increments of the quotients, and put $D = dv - eu$. Then

$$D\,z = df + ude + (u-1)e^2. \tag{4}$$

So D is a *positive* integer, even though dv and eu largely cancel, and it is small when the drops are small. Also $(q, x) = (\lfloor a/b_i \rfloor, b_i)$ lies on the curve $xM(q) = aN(q) + D$, where M and N are explicit polynomials depending on (u, v) (Lemma 3.3). A resultant-type identity shows that each such curve has $a^{o(1)}$ admissible points (Lemma 4.1). The number of triples (u, v, D) is $\ll (B/T)^2 \cdot a/T^3$, where $B = a/A$, which gives (3).

A second positive integer. Six consecutive terms carry three overlapping curvatures D, E, F and four quotient increments u, v, w, z. In Section 7 we show that the integer

$$L = \frac{S(v,w,z)\,D - S(u,v,w)\,E}{v}, \qquad S(u,v,w) = u^2v + (u-1)v^2 + uw,$$

satisfies the exact identity $qwL = J(u,v,w,z)\,E + S(u,v,w)\,F$, where J is a polynomial with positive values (Lemma 7.1). Hence $L \ge 1$, and $q \asymp a/A$ is bounded by a polynomial in the small parameters u, v, w, z, E, F. A single window of six terms with small total drop then gives

$$T^6 A^2 \le 96 \cdot 10^6\, a^3 \qquad (T \ge 10), \tag{5}$$

with no divisor function at all (Proposition 7.3).

With (5) in hand, the only remaining loss is in the drop bound (2). That loss comes from the number of divisors of $a + h$ in $[A, 2A)$ as h ranges over a short interval. Section 8 shows that any bound $\ll y\,G(x)$ for such counts over intervals $(x, x+y]$ with $y \ge x^{1/4}$ gives $P(a,b) \ll a^{3/10} G(a)^{1/5}$ (Theorem 8.3). The counts are majorised by the Erdős–Hooley function. Combining the short-sum theorem of Nair and Tenenbaum [7] with the recent mean-value bound of de la Bretèche and Tenenbaum [1] gives $G(x) = (\log\log x)^{5/2}$, and hence the following.

Theorem 1.2. *Assume the theorem of Nair–Tenenbaum stated as* (9) *and the theorem of de la Bretèche–Tenenbaum stated as* (10) *below. Then*

$$P(a,b) \ll a^{3/10}\sqrt{\log\log 3a} \qquad \text{uniformly for integers } 0 < b < a.$$

Both hypotheses are published theorems. They are hypotheses only in the sense of the formalisation: we have not formalised their proofs. Any other short-interval divisor bound can be used in their place. For example, Shiu's theorem [12] gives $\sum_{x<n\le x+y} \tau(n) \ll y \log x$ for $y \ge x^{1/4}$, and hence $P(a,b) \ll a^{3/10}(\log a)^{1/5}$; we have not formalised that either.

Structured trajectories and the Hankel hierarchy. Write $k_i = q_{i+1} - q_i$ for the quotient increments, and let $R-1$ be the number of indices with $k_i < k_{i+1}$.

Theorem 1.3. *$P(a,b) \ll a^{1/4}\sqrt{R}$ uniformly for $0 < b < a$. In particular, if the quotient increments never rise, then $P(a,b) \ll a^{1/4}$.*

The proof (Section 9) combines three facts. First, $D \geq 1$ says that d_i/k_i is strictly decreasing, so the drops strictly decrease along every stretch without a rise. Second, the six-step integer L gives an exact identity for six quotients in arithmetic progression, which forces their common difference to be large (Lemma 9.1). Third, a counting lemma for sequences in which no value occurs too often. So an orbit of length about $a^{3/10}$ must contain $\gg a^{1/10}$ rises.

Finally, both D and L fit into a single structure (Section 10). The orbit carries an exact finite model by positive measures ν_i with $\int d\nu_i = b_i/a$. The Hankel determinants $I_r(i) = a \det\bigl(\int x^{j+l}\, d\nu_i\bigr)_{j,l<r}$ are positive integers, and I_1, I_2, I_3, I_4 are b_i, $b_i - b_{i+1}$, D and L (Theorem 10.2).

Below 3/10. Our main result goes past the exponent 3/10.

Theorem 1.4. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \leq C_\varepsilon\, a^{23/77+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $23/77 = 0.298701\ldots$, a saving of $1/770$ over $3/10$. The proof (Section 11) uses the fourth Hankel invariant $I_4 = L$ as an integer, not only as a positive number. At the critical configuration L has bounded size. The labels $L(i), L(i+1), L(i+2)$ and four cofactors of an explicit 3×4 integer matrix form a *pattern* of seven integers. Two consecutive states give two linear equations and one quadratic equation in five consecutive quotient increments, and the quadratic has a positive leading coefficient. An exact identity $I_4(i)\, p_i(-q_i) = a \det(\mu_i(x^{j+l}))_{j,l<4}$, with p_i the degree-four orthogonal polynomial of the moment model, makes q_i a root of a quartic that is fixed by the pattern and the increments. So each pattern accounts for $O(H^2)$ states, where H bounds the increments. The number of patterns is controlled by Hadamard's inequality applied to the moment model.

The idea of encoding states by the coefficients of the degree-four orthogonal polynomial is due to an AI-assisted exploration (see the acknowledgements). It first gave a bound $a^{3/10-\delta}$ for a small $\delta > 0$, through the resultant identity $\mathrm{Res}(I_4 p_4, I_5 p_5) = a I_5^8$ for the orthogonal polynomials of degrees four and five, together with the dimension-growth bounds of Cluckers, Dèbes, Hendel, Nguyen and Vermeulen [2]. The two-state count of Section 11 replaces both by elementary algebra.

Formal verification. Every step of the proofs of Theorems 1.1 and 1.4 has been formalised in Lean 4 [5] using Mathlib [6]. So have Proposition 7.3, Theorem 8.3, the deduction of Theorem 1.2 from (9) and (10), Theorem 1.3 and Theorem 10.2. Section 12 describes the formalisation. The constants in this paper are the constants in the Lean proof.

What is not proved. These results are far from (1), and the method cannot reach it; see Section 13.

2. Preliminaries

2.1. **Notation.** Throughout, $a > b > 0$ are integers, $b_1 = b$, $b_{i+1} = a \bmod b_i$, and $P = P(a,b)$, so that $a > b_1 > b_2 > \cdots > b_P > 0 = b_{P+1}$. Write $q_i = \lfloor a/b_i \rfloor$, so

$$a = q_i b_i + b_{i+1}, \qquad 0 \leq b_{i+1} < b_i \qquad (1 \leq i \leq P).$$

Let $\tau(n)$ be the number of positive divisors of n, and let

$$\Delta = \Delta(a) = \max_{1 \le n \le 32a^7} \tau(n).$$

Lemma 2.1 (Quotients increase)**.** *If $b_{i+1} > 0$ then $q_{i+1} \ge q_i + 1$.*

Proof. Since $b_{i+1} < b_i$ we have $(q_i+1)b_{i+1} \le q_i b_i + b_{i+1} = a$, so $q_{i+1} = \lfloor a/b_{i+1} \rfloor \ge q_i + 1$. □

Lemma 2.2 (Divisor bound)**.** *For every integer $k \ge 1$ there is C_k with $\tau(n)^k \le C_k\, n$ for all $n \ge 1$.*

Proof. This is standard: $\tau(n)^k/n = \prod_{p^\alpha \| n} (\alpha+1)^k / p^\alpha$. Each factor with $p \ge 2^k$ is at most $(\alpha+1)^k/2^{k\alpha} \le 1$. Each of the fewer than 2^k factors with $p < 2^k$ is at most $\sup_\alpha (\alpha+1)^k/2^\alpha < \infty$. □

2.2. **Window runs.**

Definition 2.3. Let $A \ge 1$. A *window run* of length T is a sequence $x_0, x_1, \dots, x_{T-1}$ of consecutive terms of the sequence (b_i), so $x_{j+1} = a \bmod x_j$, all lying in $[A, 2A)$.

In a window run, $x_0 > x_1 > \cdots > x_{T-1}$ and $x_j \le a$.

3. The curvature of three consecutive steps

Fix four consecutive terms

$$x = b_i, \quad y = b_{i+1} = a \bmod x, \quad z = b_{i+2} = a \bmod y, \quad w = b_{i+3} = a \bmod z,$$

with $x, y, z > 0$ (and $w \ge 0$). Write

$$q = \left\lfloor \frac{a}{x} \right\rfloor, \quad u = \left\lfloor \frac{a}{y} \right\rfloor - q, \quad v = \left\lfloor \frac{a}{z} \right\rfloor - \left\lfloor \frac{a}{y} \right\rfloor, \qquad d = x - y, \quad e = y - z, \quad f = z - w.$$

By Lemma 2.1, $u, v \ge 1$, and clearly $d, e, f \ge 1$. The division equations read

$$xq + y = a, \qquad y(q+u) + z = a, \qquad z(q+u+v) + w = a. \tag{6}$$

Definition 3.1. The *curvature* of the block is $D = dv - eu$.

Lemma 3.2 (Curvature identity)**.**

$$D\,z = df + ude + (u-1)e^2.$$

In particular $D \ge 1$.

Proof. Subtracting the first two equations of (6) gives $uy = qd + e$, hence $uz = uy - ue = qd + (1-u)e$. Subtracting the last two gives $vz = (q+u)e + f$. Therefore

$$(dv - eu)z = d\big((q+u)e + f\big) - e\big(qd + (1-u)e\big) = df + ude + (u-1)e^2.$$

The right-hand side is at least $df \ge 1$, and $z > 0$, so $D > 0$. □

Equivalently, the identity is the linear combination

$$Dz - \big(df + ude + (u-1)e^2\big) = d\,E_3 + (a - qx - x - y + z)\,E_2 + (-a + (q+u)y + y)\,E_1$$

of the three division equations E_1, E_2, E_3 of (6) (each written as left $-$ right $= 0$). This is how it is checked in Lean.

Lemma 3.3 (The fiber equation)**.** *Put $M(q) = uq^2 + (u^2+u+v)q + v$ and $N(q) = uq + u^2 + v$. Then*

$$x\,M(q) = a\,N(q) + D.$$

Proof. From (6), $d = (q+1)x - a$ and $e = y - z = (q+u+1)y - a = a(q+u) - q(q+u+1)x$. Substituting these into $D = dv - eu$ gives $D = x\big(uq(q+u+1) + v(q+1)\big) - a\big(u(q+u) + v\big)$, which is the claim. □

Lemma 3.4 (Size of the parameters)**.** *If $x \le a$ then*

$$u\,yx \le a(d+e), \qquad v\,zy \le a(e+f), \qquad D\,z \le (u+1)(d+e+f)^2.$$

Proof. From $uy = qd + e$ we get $uyx = (qx)d + ex \le ad + ae$, since $qx \le a$ and $x \le a$. Similarly $vz = (q+u)e + f$ with $(q+u)y \le a$ and $y \le a$ gives the second bound. For the third, by Lemma 3.2, $Dz = df + ude + (u-1)e^2 \le df + u\,e(d+e) \le (1+u)(d+e+f)^2$. □

4. Few blocks share a triple (u, v, D)

Lemma 4.1 (Fiber count)**.** *Let $a \ge 1$, $1 \le u, v \le a$ and $1 \le D \le 2a$. Let $M(q) = uq^2 + pq + v$ with $p = u^2 + u + v$, and let $L(q) = cq + \ell$ with $c = au$ and $\ell = a(u^2+v) + D$. Then the set*

$$Q = \{q \ge 0 :\ xM(q) = L(q) \text{ for some integer } x \ge 1\}$$

has at most $\Delta = \max_{n \le 32a^7} \tau(n)$ elements.

Proof. Put $U(q) = ucq + cp - u\ell$ and $R = u\ell^2 - cp\ell + c^2 v$. Expanding shows

$$c^2 M(q) - U(q)L(q) = R \tag{7}$$

identically in q. If $q \in Q$ with witness x, substituting $L(q) = xM(q)$ into (7) gives

$$R = M(q)\big(c^2 - U(q)x\big). \tag{8}$$

We bound the sizes first. From $u, v \le a$ and $D \le 2a$ we get $\ell \le 4a^3$, $p \le 3a^2$ and $c \le a^2$, so

$$|R| \le u\ell^2 + cp\ell + c^2 v \le 16a^7 + 12a^7 + a^5 \le 32a^7, \qquad c^2 \le a^4 \le 32a^7.$$

Case $R \ne 0$. By (8), $M(q)$ is a positive divisor of $|R|$. Since $u, p \ge 1$, the map $q \mapsto M(q)$ is strictly increasing on $q \ge 0$, hence injective. Therefore $|Q| \le \tau(|R|) \le \Delta$.

Case $R = 0$. Since $M(q) > 0$, (8) gives $U(q)x = c^2$. Here $c^2 > 0$ and $x \ge 1$, so $U(q)$ is a positive divisor of c^2. Since U is linear with positive leading coefficient uc, it is injective. Therefore $|Q| \le \tau(c^2) \le \Delta$. □

The case $R = 0$ cannot be dropped: it is the case in which M and L might share a factor.

5. Counting visits to a window

In this section $x_0, \dots, x_{T-1}$ is a window run in $[A, 2A)$, and $\Delta = \max_{n \le 32a^7} \tau(n)$.

Proposition 5.1 (Drop bound; after Erdős–Shallit)**.** *If $T \ge 2$ then $T(T-1) \le 4A\Delta$.*

Proof. Let $h_j = x_j - x_{j+1} \ge 1$ for $0 \le j \le T-2$. By telescoping, $\sum_j h_j = x_0 - x_{T-1} \le (2A-1) - A = A - 1$. Call j *good* if $h_j(T-1) \le 2A$ and *bad* otherwise. Each bad j has $h_j(T-1) \ge 2A+1$, so

$$|\,\mathrm{Bad}\,|\,(2A+1) \le (T-1)\textstyle\sum_j h_j \le (T-1)(A-1).$$

If $2|\,\mathrm{Bad}\,| \ge T-1$ this would give $(T-1)(2A+1) \le 2(T-1)(A-1)$, which is false. Hence $2|\,\mathrm{Bad}\,| < T-1$, and since $|\,\mathrm{Good}\,| + |\,\mathrm{Bad}\,| = T-1$ we get $2|\,\mathrm{Good}\,| \ge T$.

For good j we have $1 \le h_j \le H := \lfloor 2A/(T-1) \rfloor$. Also $a + h_j = q_j x_j + x_{j+1} + x_j - x_{j+1} = (q_j+1)x_j$, so $x_j \mid a + h_j$. Moreover $a + h_j \le 2a \le 32a^7$. Since $j \mapsto x_j$ is injective, at most $\tau(a+h) \le \Delta$ good indices have $h_j = h$. Hence $|\,\mathrm{Good}\,| \le H\Delta$, and

$$T(T-1) \le 2|\,\mathrm{Good}\,|(T-1) \le 2\Delta\,H(T-1) \le 4A\Delta.$$ □

Proposition 5.2 (Curvature bound)**.** *If $T \geq 12$ then $T^6A^2 \leq 995328\,a^3\Delta$.*

Proof. For $0 \leq j \leq T-4$ let $s_j = x_j - x_{j+3}$ be the total drop of the block $x_j, x_{j+1}, x_{j+2}, x_{j+3}$. By telescoping,

$$\sum_{j=0}^{T-4} s_j = (x_0 + x_1 + x_2) - (x_{T-3} + x_{T-2} + x_{T-1}) \leq 3(2A-1) - 3A = 3A - 3.$$

Call j *good* if $s_j T \leq 12A$ and *bad* otherwise. Then $|\,\mathrm{Bad}\,|(12A+1) \leq T(3A-3)$. If $4|\,\mathrm{Bad}\,| \geq T$ this would give $T(12A+1) \leq 4T(3A-3)$, which is false. So $4|\,\mathrm{Bad}\,| \leq T-1$. Since $|\,\mathrm{Good}\,| = T - 3 - |\,\mathrm{Bad}\,|$ and $T \geq 12$,

$$2|\,\mathrm{Good}\,| \geq 2T - 6 - \tfrac{T-1}{2} \geq T.$$

Fix a good j. Apply Section 3 to $x = x_j$, so that $y, z, w = x_{j+1}, x_{j+2}, x_{j+3}$, all in $[A, 2A)$. Write $s = s_j = d + e + f$. Lemma 3.4 gives $uA^2 \leq uyx \leq a(d+e) \leq as$. Multiplying by T and using $sT \leq 12A$ gives $uA^2T \leq 12aA$, that is,

$$u\,AT \leq 12a, \qquad \text{and likewise} \qquad v\,AT \leq 12a.$$

Also $DA \leq Dz \leq (u+1)s^2 \leq 2us^2$. Multiplying by T^2 gives $DAT^2 \leq 2u(sT)^2 \leq 288uA^2$, so $DT^2 \leq 288uA$, and then

$$D\,T^3 \leq 288\,u\,AT \leq 3456\,a.$$

Thus the triple (u, v, D) of a good block lies in $[1, U]^2 \times [1, D^*]$, where

$$U = \lfloor 12a/(AT) \rfloor \leq a, \qquad D^* = \lfloor 3456a/T^3 \rfloor \leq 2a.$$

The inequalities $U \leq a$ and $D^* \leq 2a$ hold because $AT \geq 12$ and $T^3 \geq 1728$.

Now fix such a triple and consider the good blocks that have it. By Lemma 3.3, each one gives a point $q = q_j = \lfloor a/x_j \rfloor$ of the set Q of Lemma 4.1. The map $j \mapsto q_j$ is injective on these blocks: q_j determines $x_j = L(q_j)/M(q_j)$, and x_j determines j. By Lemma 4.1, at most Δ good blocks share a triple. So $|\,\mathrm{Good}\,| \leq U^2 D^* \Delta$, and

$$T^6A^2 = T\,(AT)^2\,T^3 \leq 2\Delta\,(U\,AT)^2\,(D^*T^3) \leq 2\Delta(12a)^2(3456a) = 995328\,a^3\Delta. \qquad \square$$

Corollary 5.3 (Window bound)**.** *If $T \geq 12$ then $T^{10} \leq 63700992\,a^3\Delta^3$.*

Proof. Using $T \leq 2(T-1)$ and the two propositions,

$$T^{10}A^2 \leq 4\,(T^6A^2)\,\big(T(T-1)\big)^2 \leq 4 \cdot 995328\,a^3\Delta \cdot (4A\Delta)^2 = 63700992\,a^3\Delta^3A^2.$$

Divide by A^2. $\square$

Remark 5.4. In terms of $B = a/A$, Propositions 5.1 and 5.2 say $T \ll a^{1/2+o(1)}B^{-1/2}$ and $T \ll a^{1/6+o(1)}B^{1/3}$. The first is better when $B > a^{2/5}$, and both equal $a^{3/10+o(1)}$ at $B = a^{2/5}$. Corollary 5.3 takes the product of the two bounds, so no case split is needed.

6. Proof of Theorem 1.1

Let $0 < b < a$. For $0 \leq k \leq \log_2 a$ let T_k be the number of indices $i \leq P$ with $2^k \leq b_i < 2^{k+1}$. Every b_i with $i \leq P$ lies in $[1, a)$, so $P = \sum_{0 \leq k \leq \log_2 a} T_k$. Since (b_i) is decreasing, the indices counted by T_k are consecutive. So when $T_k > 0$ they form a window run in $[A, 2A)$ with $A = 2^k$.

Let $\varepsilon > 0$. Choose an integer $m > 7/(5\varepsilon)$, and let C_{3m} be the constant of Lemma 2.2 with $k = 3m$. The maximum Δ is attained at some $n_0 \leq 32a^7$, so $\Delta^{3m} \leq C_{3m} \cdot 32a^7$. If $T_k \geq 12$, Corollary 5.3 gives

$$T_k^{10m} \leq 63700992^m\,a^{3m}\,\Delta^{3m} \leq K\,a^{3m+7}, \qquad K = 63700992^m \cdot C_{3m} \cdot 32,$$

and hence

$$T_k \le K^{1/(10m)}\, a^{3/10+7/(10m)} \le K^{1/(10m)}\, a^{3/10+\varepsilon/2}.$$

If $T_k \le 11$ this bound holds with 11 in place of $K^{1/(10m)}$. Finally $\lfloor \log_2 a \rfloor + 1 \le C'_\varepsilon a^{\varepsilon/2}$. Summing over k,

$$P(a,b) \le (\lfloor \log_2 a \rfloor + 1)\big(11 + K^{1/(10m)}\big) a^{3/10+\varepsilon/2} \le C_\varepsilon\, a^{3/10+\varepsilon}. \qquad \square$$

7. Six consecutive steps

Fix six consecutive terms $x_0 > x_1 > \cdots > x_5 \ge 0$ of the sequence (b_i) with $x_4 > 0$, and write their quotients as

$$\lfloor a/x_0 \rfloor = q, \quad \lfloor a/x_1 \rfloor = q+u, \quad \lfloor a/x_2 \rfloor = q+u+v,$$
$$\lfloor a/x_3 \rfloor = q+u+v+w, \quad \lfloor a/x_4 \rfloor = q+u+v+w+z.$$

(In this section w is a quotient increment, not a term of the sequence.) By Lemma 2.1, $u, v, w, z \ge 1$. Let $f_k = x_k - x_{k+1} \ge 1$ be the drops, and let

$$D = f_0 v - f_1 u, \qquad E = f_1 w - f_2 v, \qquad F = f_2 z - f_3 w$$

be the curvatures of the blocks starting at x_0, x_1 and x_2. Write E_k for the division equation $x_k q_k + x_{k+1} - a = 0$, where $q_k = \lfloor a/x_k \rfloor$. Put

$$S(u,v,w) = u^2 v + (u-1)v^2 + uw,$$
$$J(u,v,w,z) = \big(u^2 + (u-1)v\big)z + w\big(u^3 + uv(2u-1) + (u-1)v^2\big) + w^2\big((u-1)(u+v)+1\big),$$
$$L = (vw + w^2 + z)\, D - \big(u^2 + (u-1)v\big)E - w\,(wf_0 - uf_2).$$

Lemma 7.1 (The six-step identity)**.** (1) $S(v,w,z)\, D - S(u,v,w)\, E = v\, L$.
(2) $q\, w\, L = J(u,v,w,z)\, E + S(u,v,w)\, F$.

Proof. Part (1) is a polynomial identity: since $S(v,w,z) \equiv -w^2$ and $S(u,v,w) \equiv uw \pmod v$, and $wD + uE = v(wf_0 - uf_2)$, the difference is divisible by v, and expanding gives L. Part (2) holds modulo the division equations. Explicitly, $qwL - JE - S(u,v,w)F$ equals

$$w(v^2w + vw^2 + vz - w^2)\, E_0 - w(u^2w + 2uvw + uw^2 + uz + v^2w + vw^2 - vw + vz - w^2)\, E_1$$
$$+ w(u^2v + u^2w + uv^2 + 2uvw + uw^2 + uw + uz - v^2 - vw)\, E_2 - w(u^2v + uv^2 + uw - v^2)\, E_3.$$

This is an identity of polynomials in a, q, the x_k and the increments, checked by expansion. $\square$

Corollary 7.2. *Suppose $q \ge 1$ and $u, v, w, z \le m$. Then $E, F \ge 1$, $L \ge 1$, and*

$$q \le 9m^3 E + 3m^3 F.$$

Proof. Lemma 3.2, applied to the blocks starting at x_1 and x_2, gives $E, F \ge 1$. Clearly $J > 0$ and $S \ge 0$, so by Lemma 7.1(2) $qwL > 0$. Since L is an integer, $L \ge 1$ and $qw \le JE + SF$. Bounding each monomial gives $J \le 9m^3 w$ and $S(u,v,w) \le 3m^3 w$. Dividing by w gives the claim. $\square$

So the quotient q, which is about a/A, is controlled by the small parameters of a single block of six terms. No counting over many blocks is needed.

Proposition 7.3 (Six-step window bound)**.** *For a window run of length $T \ge 10$,*

$$T^6 A^2 \le 96 \cdot 10^6\, a^3.$$

Proof. For $0 \le j \le T-6$ let $s_j = x_j - x_{j+5}$. By telescoping, $\sum_{j \le T-6} s_j \le 5(2A-1) - 5A = 5A-5$. So some j has $s_j(T-5) \le 5A-5$, and since $T \le 2(T-5)$, we get $s := s_j$ with $sT \le 10A$. Apply Corollary 7.2 to $x_j, \dots, x_{j+5}$, all of which lie in $[A, 2A)$, with $m = \max(u, v, w, z)$.

By Lemma 3.4 at $x_j, \dots, x_{j+3}$, each increment satisfies (increment) $\cdot A^2 \le as$. Multiplying by T gives $mAT \le 10a$. By Lemma 3.4 again, $EA \le Ex_{j+3} \le (v+1)s^2 \le 2ms^2$, and likewise $FA \le 2ms^2$. Hence $qA \le 24m^4s^2$. Also $a = qx_j + x_{j+1} < (q+1)x_j < 2A(q+1) \le 4Aq$. Therefore

$$qA \cdot A^4T^6 \le 24\,(mAT)^4(sT)^2 \le 24 \cdot 10^6\, a^4A^2,$$

so $qA^3T^6 \le 24 \cdot 10^6 a^4$, and $aA^2T^6 \le 4qA^3T^6 \le 96 \cdot 10^6\, a^4$. □

Remark 7.4. In terms of $B = a/A$, Proposition 7.3 reads $T \ll a^{1/6}B^{1/3}$. This is Proposition 5.2 without the factor $\Delta^{1/6}$. Combined with Proposition 5.1 as in Corollary 5.3, it gives $T^{10} \ll a^3\Delta^2$ and another proof of Theorem 1.1. The Lean proof of Theorem 1.1 uses Corollary 5.3.

8. Divisors over short intervals

For $A \ge 1$ let $\tau_A(n) = \#\{d \mid n : \ A \le d < 2A\}$.

Proposition 8.1 (Drop count)**.** *For a window run of length $T \ge 2$ and any $Y \ge \lfloor 2A/(T-1) \rfloor$,*

$$T \le 2 \sum_{a < n \le a+Y} \tau_A(n).$$

Proof. This is the proof of Proposition 5.1, stopped before the divisor bound. The good indices j have $1 \le h_j \le \lfloor 2A/(T-1) \rfloor$, and $j \mapsto x_j$ maps those with $h_j = h$ injectively into the divisors of $a+h$ in $[A, 2A)$. □

Definition 8.2. Let $G\colon \mathbb{N} \to \mathbb{R}$. We say that $\mathrm{SD}(G)$ holds if there are C, x_0 such that for all integers $x \ge x_0$, $x^{1/4} \le y \le x$ and $A \ge 1$,

$$\sum_{x < n \le x+y} \tau_A(n) \le C\, y\, G(x).$$

Theorem 8.3. *Suppose* $\mathrm{SD}(G)$ *holds, $G \ge 1$, and $G(x) \ll x^{1/40}$. Then $P(a,b) \ll a^{3/10}\, G(a)^{1/5}$ uniformly for $0 < b < a$.*

Proof. Small a are handled by $P(a,b) \le b < a$. Fix a dyadic window $[A, 2A)$, $A = 2^k$, with $T \ge 20$ visits. Proposition 7.3 gives $T \le \alpha A^{-1/3}$, where $\alpha = (96 \cdot 10^6)^{1/6} a^{1/2}$. Take $Y = \max(\lfloor 2A/(T-1) \rfloor, \lceil a^{1/4} \rceil) \le a$ in Proposition 8.1. With $H = \lfloor 2A/(T-1) \rfloor$, $\mathrm{SD}(G)$ gives $T \le 2CG(a)(H + a^{1/4} + 1)$. Multiplying by $T-1$, using $H(T-1) \le 2A$, and solving the quadratic gives $T \le \gamma + \beta A^{1/2}$, where $\beta = 4\sqrt{CG(a)}$ and $\gamma = 4CG(a)(a^{1/4}+1)$. So every window satisfies

$$T_k \le 19 + \gamma + \min\bigl(\alpha\, 2^{-k/3},\ \beta\, 2^{k/2}\bigr).$$

Let $M = \alpha^{3/5}\beta^{2/5}$. Since $X^3Y^2 = M^5$ for $X = \alpha 2^{-k/3}$ and $Y = \beta 2^{k/2}$, for each k either $Y \le M$ or $X < M$. So the minimum is at most the sum of a geometric series with ratio $\sqrt{2}$ capped at M, plus one with ratio $2^{-1/3}$ capped at M. This gives

$$\sum_k \min\bigl(\alpha\, 2^{-k/3},\ \beta\, 2^{k/2}\bigr) \le 9\, \alpha^{3/5}\beta^{2/5} \ll a^{3/10} G(a)^{1/5},$$

with no factor $\log a$. The remaining terms contribute $(\log_2 a + 1)(19 + \gamma) \ll a^{1/100} \cdot a^{1/4} G(a)$. This is $\ll a^{3/10} G(a)^{1/5}$, because $G(a)^{4/5} \ll a^{1/50}$. □

The Erdős–Hooley function. Let $\Delta_{\mathrm{H}}(n) = \max_{u\in\mathbb{R}} \#\{d \mid n : \ e^u < d \le e^{u+1}\}$. Taking $e^{u+1} = 2A$, and using $e > 2$, gives $\tau_A(n) \le \Delta_{\mathrm{H}}(n)$. We use two published theorems.

(NT) Nair and Tenenbaum [7] prove a general upper bound for short sums of functions F satisfying $F(mn) \le \min(A^{\Omega(m)}, Bm^{\varepsilon})F(n)$ for $(m,n) = 1$. They note that Δ_{H} is such a function. In their introduction they record the consequence, for $Q(X) = X$ and $t = 1$, that for x large and $x^{\varepsilon} \le y \le x$

$$\sum_{x<n\le x+y} \Delta_{\mathrm{H}}(n) \ll_{\varepsilon} \frac{y}{\log x} \sum_{n\le x} \frac{\Delta_{\mathrm{H}}(n)}{n}. \tag{9}$$

We use it with $\varepsilon = 1/4$.

(BT) de la Bretèche and Tenenbaum [1, Theorem 1.1] prove that for large x

$$\sum_{n\le x} \Delta_{\mathrm{H}}(n) \ll x\,(\log\log x)^{5/2}. \tag{10}$$

Lemma 8.4. *Assume* (10)*, and let* $\ell(x) = \max(1, \log\log x)$*. Then* $\sum_{n\le x} \Delta_{\mathrm{H}}(n)/n \ll \log x \cdot \ell(x)^{5/2}$ *for large* x*.*

Proof. Split n into dyadic blocks $2^j \le n < 2^{j+1} \le 2x$. The terms with n below the threshold of (10) contribute $O(1)$, since $\Delta_{\mathrm{H}}(n) \le n$. Each remaining block contributes at most $2^{-j}\sum_{n\le 2^{j+1}} \Delta_{\mathrm{H}}(n) \ll \ell(2x)^{5/2}$, and $\ell(2x) \le 2\ell(x)$. There are at most $\log_2 x + 1 \le 3\log x$ blocks. □

Proof of Theorem 1.2. By (9) and Lemma 8.4, $\sum_{x<n\le x+y} \tau_A(n) \le \sum_{x<n\le x+y} \Delta_{\mathrm{H}}(n) \ll y\,\ell(x)^{5/2}$ for $x^{1/4} \le y \le x$. So $\mathrm{SD}(G)$ holds with $G = \ell^{5/2}$, which satisfies $G \ge 1$ and $G(x) \ll x^{1/40}$. Theorem 8.3 gives $P(a,b) \ll a^{3/10}\ell(a)^{1/2} \ll a^{3/10}\sqrt{\log\log 3a}$. □

9. Rises in the quotient increments

Throughout this section $k_i = q_{i+1} - q_i \ge 1$, $d_i = b_i - b_{i+1} \ge 1$, and a *rise* is an index with $k_i < k_{i+1}$.

Lemma 9.1 (Six quotients in arithmetic progression)**.** *Let* $q, h \ge 1$ *and* $x_0 > x_1 > \cdots > x_6 \ge 0$ *satisfy* $x_j(q+jh) + x_{j+1} = a$ *for* $0 \le j \le 5$*. Then* $q^4 \le 324\,h^3 a$*.*

Proof. Let $f_j = x_j - x_{j+1}$, and let $E = (f_1 - f_2)h$ and $F = (f_2 - f_3)h$ be the curvatures of the blocks starting at x_1 and x_2 (all their increments equal h). With $U = f_0 - 2f_1 + f_2$, the division equations e_j give the exact identity

$$qhU = (3h-1)E + F, \qquad \text{namely} \qquad qhU - (3h-1)E - F = h(e_0 - 3e_1 + 3e_2 - e_3).$$

This is the case $u = v = w = z = h$ of Lemma 7.1, where $L = 2h^3U$. By Lemma 3.2, $E, F \ge 1$, so the integer U is at least 1.

Now bound E and F. Subtracting consecutive division equations gives $q_j f_j \le h x_{j+1}$, where $q_j = q + jh$. Since $x_{j+1}q_{j+1} \le a$, this yields $q^2 f_j \le ha$ for $1 \le j \le 4$. By Lemma 3.4, $Ex_3 \le (h+1)(f_1+f_2+f_3)^2 \le 9h^2(h+1)a^2/q^4$. Also $a < x_3(q+3h+1)$, so $Eq^4 \le 9h^2(h+1)(q+3h+1)a$, and likewise $Fq^4 \le 9h^2(h+1)(q+4h+1)a$. Hence

$$q^5 h \le q^4\big((3h-1)E + F\big) \le 27h^3(h+1)(q+4h+1)a.$$

If $q \ge h$ this gives $q^4 \le 324h^3a$. If $q < h$ then $q^4 < h^4 \le h^3 a$. □

Remark 9.2. The source of this section proved a version of Lemma 9.1 for seven quotients by an analytic argument: it represents a periodic continuation of the orbit by integrals of positive functions, and compares finite differences. The identity above gives an algebraic proof with one fewer quotient.

Lemma 9.3. *Whenever $b_{i+2} > 0$, $d_{i+1}k_i < d_ik_{i+1}$. In particular, if $k_{i+1} \le k_i$ then $d_{i+1} < d_i$.*

Proof. This is $D = d_ik_{i+1} - d_{i+1}k_i \ge 1$ (Lemma 3.2). If $k_{i+1} \le k_i$ and $d_{i+1} \ge d_i$, then $d_ik_{i+1} \le d_{i+1}k_i$, which is a contradiction. □

Lemma 9.4. *Let $c_j \ge 1$ be integers, $j \in J$, such that no value occurs more than M times. Then $|J|^2 \le 2M \sum_{j\in J} c_j$.*

Proof. Induct on $|J|$. Let c_{j_0} be the largest value. The values lie in $[1, c_{j_0}]$, so $|J| \le Mc_{j_0}$. Removing j_0 and using $|J|^2 = (|J|-1)^2 + 2|J| - 1$ completes the induction. □

Proposition 9.5. *Let $x_0, \dots, x_{T-1}$ be a window run in $[A, 2A)$ with $T \ge 2$, and let ρ be the number of rises among its $T-1$ increments. Then:*

1. *$(T-1)^2 \le 2(\rho+1)(A-1)$;*
2. *$T - 1 \le \lfloor a/A \rfloor$;*
3. *if $6A \le a$, then $T - 1 = \ell + s$ with $\ell^2 \le 8(\rho+1)\lfloor a/A \rfloor$ and $s^3 \le 26244\,A$.*

Proof. Split the increments into *runs*: maximal stretches containing no rise. There are at most $\rho + 1$ runs. Map each index to the end of its run.

(1) By Lemma 9.3, the drops strictly decrease along a run, so each value of d_j occurs at most once per run, hence at most $\rho + 1$ times. Lemma 9.4 and $\sum_j d_j = x_0 - x_{T-1} \le A - 1$ give the claim.

(2) The increments are ≥ 1, and they sum to $q_{T-1} - q_0 \le \lfloor a/A \rfloor$.

(3) Call an increment h *low* if $26244h^3A^4 < a^3$. Five consecutive equal increments h inside the run give six quotients in arithmetic progression. Lemma 9.1 then gives $q^4 \le 324h^3a$. On the other hand, $a < 3Aq$, which follows from $a = qx_0 + x_1$, $x_1 < x_0 < 2A$ and $6A \le a$. Together these give $a^3 \le 26244h^3A^4$, so h is not low. Within a run the increments are nonincreasing, so equal values are consecutive. Hence each low value occurs at most 4 times per run, and at most $4(\rho+1)$ times in all. Lemma 9.4 bounds the number ℓ of low increments by $\ell^2 \le 8(\rho+1)\sum_j k_j \le 8(\rho+1)\lfloor a/A \rfloor$. Let s be the number of high increments, with smallest value h_*. Then $sh_* \le \lfloor a/A \rfloor$ and $a^3 \le 26244h_*^3A^4$, so $s^3a^3 \le 26244A^4(a/A)^3 \le 26244Aa^3$. □

Proof of Theorem 1.3. The rises inside a window are rises of the orbit, so $\rho+1 \le R$. Combining the three parts of Proposition 9.5, as in the proof of Theorem 8.3, gives for the window $[2^k, 2^{k+1})$

$$T_k \le 1 + \sqrt{R}\,\min\bigl(\sqrt{8a}\,2^{-k/2},\ \sqrt{2}\,2^{k/2}\bigr) + \min\bigl(a\,2^{-k},\ 30\cdot 2^{k/3}\bigr).$$

The geometric-sum lemma, in its general form $\sum_k \min(\alpha 2^{-sk}, \beta 2^{tk}) \ll_{s,t} \alpha^{t/(s+t)}\beta^{s/(s+t)}$, gives $\sum_k \min(\sqrt{8a}\,2^{-k/2}, \sqrt{2}\,2^{k/2}) \ll (\sqrt{8a}\sqrt{2})^{1/2} = 2a^{1/4}$. It also gives $\sum_k \min(a2^{-k}, 30\cdot 2^{k/3}) \ll a^{1/4}$. Since $\log_2 a + 1 \ll a^{1/4}$, summing over k gives the theorem. □

Remark 9.6. By Theorem 1.3, an orbit with $P(a,b) \asymp a^{3/10}$ must have $R \gg a^{1/10}$. Proposition 9.5(3) says more: inside the critical window $A \asymp a^{3/5}$, a run of $T \asymp a^{3/10}$ remainders must contain $\gg a^{1/5}$ rises.

10. The Hankel hierarchy

Let $b_0 > b_1 > \cdots > b_{N-1} > 0$ be the positive part of an orbit (indexed from 0, so $N = P(a,b)$), with quotients q_i. Put $p_i(j) = q_{i+j} - q_i$ for $j < N-i$. Define weights $w_i(j)$ backwards by $w_{N-1} = \delta_0$ and

$$w_i(0) = 1 - \sum_j \frac{w_{i+1}(j)}{p_i(j+1)}, \qquad w_i(j+1) = \frac{w_{i+1}(j)}{p_i(j+1)}.$$

Let μ_i be the measure with mass $w_i(j)$ at $p_i(j)$, and let $\nu_i = \mu_i/(q_i + x)$. Write $h_i(s) = \int x^s \, d\nu_i = \sum_j w_i(j) p_i(j)^s / q_{i+j}$.

Lemma 10.1. (1) *The weights are positive and sum to* 1.
(2) $h_i(0) = b_i/a$, *that is,* $b_i/a = \mathbb{E}\big[1/(q_i + X_i)\big]$ *for* $X_i \sim \mu_i$.
(3) $h_i(s+1) = \sum_{t \le s} \binom{s}{t} k_i^{s-t} h_{i+1}(t)$ *for* $i + 1 < N$.
(4) $h_i(s) = (-q_i)^s b_i/a + z$ *with* $z \in \mathbb{Z}$.

Proof. (1) Since $p_i(j+1) \ge 1$, the sum defining $w_i(0)$ is at most 1. It is strictly less: if $N-i-1 \ge 2$, the term $j = 1$ has $p_i(2) \ge 2$; if $i = N-2$, the last gap satisfies $q_{N-1} \ge q_{N-2} + 2$. (2) Since $p_i(j+1) = q_{i+j+1} - q_i$, the definitions give $q_i h_i(0) + h_{i+1}(0) = 1$, which is the normalised division equation; and $h_{N-1}(0) = 1/q_{N-1} = b_{N-1}/a$. (3) Use $p_i(j+1) = k_i + p_{i+1}(j)$ and the binomial theorem. (4) Induct backwards on i using (3), $b_{i+1}/a = 1 - q_i b_i/a$ and $k_i - q_{i+1} = -q_i$. □

Theorem 10.2. *For* $1 \le r \le N - i$, *the number* $I_r(i) = a \det\big(h_i(j+l)\big)_{0 \le j,l < r}$ *is a positive integer. Moreover* $I_1(i) = b_i$, $I_2(i) = d_i$ *(if* $i+1 < N$*),* $I_3(i) = D_i$ *(if* $i+3 < N$*), and* $I_4(i) = L_i$ *(if* $i+5 < N$*), where* D_i *and* L_i *are the curvature and the six-step integer of the blocks starting at* b_i.

Proof. Integrality. By Lemma 10.1(4), $h_i(s+1) + q_i h_i(s) \in \mathbb{Z}$. Multiply the Hankel matrix on the left by the unit lower triangular matrix that adds q_i times row $j-1$ to row j. This makes rows $1, \dots, r-1$ integral and leaves the determinant unchanged. Expand along row 0, whose entries are $(-q_i)^l b_i/a + (\text{integer})$; then $a \det$ is an integer.

Positivity. The Hankel matrix is the Gram matrix of $1, x, \dots, x^{r-1}$ in $L^2(\nu_i)$. Its quadratic form is $\sum_j w_i(j) q_{i+j}^{-1} \big(\sum_l c_l p_i(j)^l\big)^2$. If this vanishes, the polynomial $\sum_l c_l x^l$ vanishes at the $N - i \ge r$ distinct points $p_i(j)$, so it is zero, by the invertibility of the Vandermonde matrix. So the matrix is positive definite.

The first four. By Lemma 10.1(2),(3), the entries $h_i(s)$, $s \le 2r-2$, are explicit polynomials in $\beta_j = b_{i+j}/a$ and $k_i, \dots, k_{i+2r-3}$. The claimed values are linear in the b_{i+j}. So each identity holds modulo the relations $q_{i+j}\beta_j + \beta_{j+1} = 1$, with explicit certificates of at most 156 terms per coefficient. □

Remark 10.3. So the positivity of D and L, proved earlier by explicit identities, is an instance of the positivity of Gram determinants. The integrality of L (Lemma 7.1(1)) is an instance of the integrality of I_4. A priori $I_r(i)$ involves the moments $h_i(s)$ for $s \le 2r-2$, and hence $b_i, \dots, b_{i+2r-2}$. In fact $I_3(i)$ depends only on b_i, b_{i+1}, b_{i+2} and q_i, q_{i+1}, q_{i+2}. Similarly $I_4(i)$ depends only on $b_i, \dots, b_{i+3}$ and $q_i, \dots, q_{i+4}$: the later remainders cancel.

11. Breaking the exponent 3/10

In this section we prove Theorem 1.4. We keep the notation of Section 10: $b_0 > b_1 > \cdots > b_{N-1} > 0$ is the positive part of the orbit, $N = P(a,b)$, and $q_i = \lfloor a/b_i \rfloor$.

11.1. **Heuristics.** The exponent 3/10 of Theorem 1.1 comes from a single balanced configuration: $T \asymp a^{3/10}$ remainders in a window $[A, 2A)$ with $A \asymp a^{3/5}$, quotients $q \asymp a^{2/5}$ and typical increments $k \asymp a^{1/10}$. At this scale the four-term Hankel invariant $I_4(i) = L(i)$ has size $\asymp aH^6/q^4 \asymp 1$, where H bounds the increments. The idea is that a positive integer of bounded size carries information. The labels $I_4(i), I_4(i+1), I_4(i+2)$, together with four further integers of the same size, determine a *pattern*. States with a given pattern satisfy two linear equations and one quadratic equation in five consecutive increments. So a pattern accounts for at most $O(H^2)$ increment tuples, and each tuple for at most four states. At the balanced point this gives $T \ll H^2 \asymp a^{1/5}$, a large saving. Moving away from the balanced point costs a power of the pattern size, and optimising gives the exponent 23/77.

11.2. **The pattern matrix.** Fix a state i with $i + 6 < N$ and write, for $m \geq 0$,

$$k_m = q_{i+m+1} - q_{i+m}, \qquad d_m = b_{i+m} - b_{i+m+1},$$
$$s_m = k_0 + \cdots + k_{m-1}, \qquad r_m = k_m(s_m + s_{m+1}) + k_{m+1}.$$

So $s_m = q_{i+m} - q_i$, $r_0 = k_0^2 + k_1$, $r_1 = k_1(2k_0 + k_1) + k_2$, and so on. Consider the 3×4 integer matrix

$$\mathcal{M}_i = \begin{pmatrix} d_0 & d_1 & d_2 & d_3 \\ k_0 & k_1 & k_2 & k_3 \\ r_0 & r_1 & r_2 & r_3 \end{pmatrix},$$

and let $U_m(i) = (-1)^{m+1} \det(\mathcal{M}_i \text{ with column } m \text{ deleted})$ for $0 \leq m \leq 3$.

Lemma 11.1. (1) *$U_3(i) = L(i) = I_4(i)$ and $U_0(i) = -L(i+1) = -I_4(i+1)$, where L is the six-step integer of Lemma 7.1.*
(2) *$\sum_m U_m(i)\, d_m = \sum_m U_m(i)\, k_m = \sum_m U_m(i)\, r_m = 0$.*

Proof. (2) is the Laplace expansion of a 4×4 determinant with a repeated row. For (1), the identity $\det(\text{columns } 0, 1, 2) = L(i)$ holds as a polynomial identity in $d_0, d_1, d_2, k_0, \ldots, k_3$, by direct expansion; and $L(i) = I_4(i)$ by Theorem 10.2. Passing from i to $i + 1$ shifts the d- and k-rows by one column, and, since $s_m(i+1) = s_{m+1}(i) - k_0$, it replaces r_m by $r_{m+1} - 2k_0k_{m+1}$. This row operation does not change the determinant, so $\det(\text{columns } 1, 2, 3) = L(i+1)$. □

Remark 11.2. The vector $C_m = U_m/U_3$ has a meaning: $p_i = \sum_{m \leq 3} C_m(F_{m+1} - F_m)$, with Newton polynomials $F_m(x) = \prod_{l<m}(x - s_l)$, is the monic orthogonal polynomial of degree four for the measure ν_i of Section 10. The three rows of $\mathcal{M}_i$ encode $\nu_i(p_i) = 0$ and $\mu_i(xp_i) = \mu_i(x^2p_i) = 0$, and $\nu_i(xf) = \mu_i(f) - q_i\nu_i(f)$ turns these into $\nu_i(x^jp_i) = 0$ for $j \leq 3$. In particular $U_m(i)/I_4(i)$ are the partial sums of the Newton coefficients of p_i, and $c_0 = -C_0 = I_4(i+1)/I_4(i)$. None of this is used below.

The *pattern* of i is the vector of seven integers

$$\theta(i) = \big(L(i),\, L(i+1),\, L(i+2),\, U_1(i),\, U_2(i),\, U_1(i+1),\, U_2(i+1)\big),$$

and its *gaps* are $\kappa(i) = (k_0, \ldots, k_4)$. By Lemma 11.1, $\theta(i)$ determines $U(i)$ and $U(i+1)$ completely.

11.3. **Three consequences.**

Lemma 11.3 (The reduced quadratic)**.** *Write $\theta(i) = (N_0, N_1, N_2, u_1, u_2, v_1, v_2)$ and $\kappa(i) = (k_0, \ldots, k_4)$. Then*

$$N_1N_0(N_1 + N_0)\, k_3^2 + \beta\, k_3 + \gamma = 0,$$

where β and γ are polynomials in $\theta(i)$, k_1 and k_2 alone.

Proof. Lemma 11.1(2) for the k-row at i and at $i+1$ gives

$$N_1k_0 = u_1k_1 + u_2k_2 + N_0k_3, \qquad N_1k_4 = N_2k_1 - v_1k_2 - v_2k_3.$$

Substitute these into $N_1^2\sum_m U_m(i)r_m = 0$. Since k_0 enters the r_m at most quadratically and k_4 only linearly, the result is a polynomial in (θ, k_1, k_2, k_3). The coefficient of k_3^2 comes from $-N_1\cdot N_1^2k_0^2$, from $N_0\cdot N_1^2\cdot 2k_0k_3$ and from $N_0\cdot N_1^2k_3^2$, and it equals $-N_1N_0^2+2N_1N_0^2+N_1^2N_0 = N_1N_0(N_1+N_0)$. Explicitly $\beta = 2N_1^2N_0(k_1+k_2)+N_1^2u_2+2N_1N_0(u_1k_1+u_2k_2)-N_1N_0v_2$, and γ is given in the Lean file `TwoState.lean`. □

Lemma 11.4 (The multiplicity identity)**.** *Let $F_m(Y) = \prod_{l<m}(Y-s_l)$ and $\mathcal{Q}_i(Y) = \sum_{m=0}^{3} U_m(i)\,F_m(Y)\,(Y-s_m-1)$. Then*

$$\mathcal{Q}_i(-q_i) = a\,H_4(k_0,\dots,k_4),$$

where H_4 is an explicit polynomial with 43 terms, independent of i. As a polynomial in q, $\mathcal{Q}_i(-q)$ has degree four and leading coefficient $U_3(i) = L(i) \ge 1$.

Proof. The division equations read $b_{i+m+1} = a - (q_i+s_m)b_{i+m}$. Use them to eliminate $b_{i+1},\dots,b_{i+4}$ from both sides. The difference becomes the zero polynomial in $a, q_i, b_i, k_0,\dots,k_4$; explicit certificates are in the Lean proof. The polynomial H_4 is the Hankel determinant $\det\big(\mu_i(x^{j+l})\big)_{j,l<4}$, and the identity is $I_4(i)\,p_i(-q_i) = a\det\big(\mu_i(x^{j+l})\big)$. This follows from the determinant formula for p_i, since $\nu_i(x^{s+1}) + q_i\nu_i(x^s) = \mu_i(x^s)$; we do not need this interpretation. The leading coefficient comes from the term $m=3$. □

Lemma 11.5 (Labels)**.** *Let $i+6<N$ and put*

$$G(k_0,\dots,k_4) = (k_0+1)\big(k_1s_2+s_2+k_1+k_2+1\big)\big(k_2(k_1+k_2)s_3+e_3\big),$$

where $e_3 = s_3(k_1+k_2)+s_3k_2+(k_1+k_2)k_2+(k_0+2k_1+3k_2)(k_3+1)+k_3^2+2k_3+k_4+1$. Then $L(i)\,q_i^4 \le a\,G(k_0,\dots,k_4)$. If moreover $b_i < 2A$, then $L(i)\,a^3 \le 256A^4\,G(k_0,\dots,k_4)$.

Proof. The Newton polynomials $F_0,\dots,F_3$ are monic of degrees $0,\dots,3$. So the matrix $\big(\nu_i(F_jF_l)\big)$ is $UHU^{\mathsf T}$, where H is the Hankel matrix of ν_i and U is unit lower triangular. Hence it has determinant $I_4(i)/a$, and it is positive definite, as in the proof of Theorem 10.2. By Hadamard's inequality,

$$\frac{I_4(i)}{a} \le \prod_{j<4}\nu_i(F_j^2) \le \prod_{j<4}\frac{\mu_i(F_j^2)}{q_i},$$

since $\nu_i = \mu_i/(q_i+x) \le \mu_i/q_i$ on the nonnegative support. By the shift rule $\mu_i(F_jg) = \mu_{i+j}\big(g(x+s_j)\big)$ and the moments $\mu_m(1)=\mu_m(x)=1$, $\mu_m(x^2)=k_m+1$, $\mu_m(x^3)=k_m^2+2k_m+k_{m+1}+1$, the product equals G. For the second claim, $a = q_ib_i+b_{i+1} < 2A(q_i+1) \le 4Aq_i$, so $L(i)a^4 \le 256A^4L(i)q_i^4 \le 256A^4aG$. □

11.4. **The size of a pattern.**

Lemma 11.6. *Let $1 \le k_0,\dots,k_5 \le H$ and $A \ge 1$. Suppose $N_0 = L(i)$ and $N_1 = L(i+1)$ satisfy $N_0a^3 \le 256A^4G(k_0,\dots,k_4)$ and $N_1a^3 \le 256A^4G(k_1,\dots,k_5)$, and that the* scale condition

$$237568\,A^4H^5 \le a^3 \tag{11}$$

holds. Then $L(i)$, $L(i+1)$, $|U_1(i)|$ and $|U_2(i)|$ are all at most $172032\,A^4H^6/a^3$.

Proof. Put $V = k_1k_2(k_1+k_2)$ and $\Delta_4 = k_0 \cdot k_1s_2 \cdot k_2(k_1+k_2)s_3$. Expanding G and bounding each of the three error terms by $56H^5$, $170H^5$ and $238H^5$ gives $G(k_0,\dots,k_4) \le \Delta_4 + 464H^5$. Also $\Delta_4 \le 6H^3V$, because $\Delta_4 = k_0s_2s_3 \cdot V$. Similarly $G(k_1,\dots,k_5) \le 6H^3V + 464H^5$, because the corresponding product is $V \cdot k_3(k_2+k_3)(k_1+k_2+k_3)$. By (11) and $N \ge 1$, $256A^4 \cdot 464H^5 \le Na^3/2$, so

$$N_0a^3,\ N_1a^3 \le 3072\,A^4H^3V.$$

Since $N_0 \ge 1$, this and (11) give $V \ge 77H^2$.

Now solve for $(u_1,u_2) = (U_1(i),U_2(i))$ from the k- and r-relations of Lemma 11.1. These read $u_1k_1 + u_2k_2 = N_1k_0 - N_0k_3$ and $u_1r_1 + u_2r_2 = N_1r_0 - N_0r_3$. The determinant is

$$\delta = k_1r_2 - k_2r_1 = k_1k_2(k_1+k_2) + k_1k_3 - k_2^2 \ \ge\ V - H^2 \ \ge\ V/2.$$

Using $r_m \le (2m+2)H^2$, Cramer's rule gives $|u_j|\delta \le 14(N_0+N_1)H^3$, hence

$$|u_j|\,Va^3 \le 28H^3(N_0+N_1)a^3 \le 28H^3 \cdot 6144A^4H^3V,$$

that is, $|u_j|a^3 \le 172032A^4H^6$. Finally $N_ja^3 \le 3072A^4H^3V \le 6144A^4H^6$, since $V \le 2H^3$. □

The point of the argument is in the first paragraph. The two adjacent labels share the Vandermonde factor V, so the ratio $L(i+1)/L(i)$ never enters the bound.

11.5. **The counting lemma.**

Proposition 11.7. *Let $\mathcal{G}$ be a set of states i with $i+8 < N$. Suppose that for each $i \in \mathcal{G}$ all entries of $\theta(i)$ have absolute value at most M, and all entries of $\kappa(i)$ lie in $[1,H]$. Then $|\mathcal{G}| \le 8(2M+1)^7H^2$.*

Proof. (i) At most four states per (θ,κ). If $\theta(i) = \theta$ and $\kappa(i) = \kappa$, then by Lemma 11.4 q_i is a root of the polynomial $\mathcal{Q}(-q) - aH_4(\kappa)$. This polynomial depends only on (θ,κ), and it has degree four and leading coefficient $\theta_0 \ge 1$. Since q_i is strictly increasing in i, there are at most four such states.

(ii) At most two values of κ per (θ,k_1,k_2). By Lemma 11.3, k_3 is a root of a quadratic whose coefficients depend only on (θ,k_1,k_2). The leading coefficient is $N_1N_0(N_1+N_0) > 0$. The two linear relations then determine k_0 and k_4, because $N_1 \ne 0$.

(iii) There are at most $(2M+1)^7H^2$ values of (θ,k_1,k_2). □

11.6. **The window bound.**

Proposition 11.8. *Let $b_s,\dots,b_{s+T-1}$ be a window run in $[A,2A)$ with $T \ge 36$, and put $H = \lfloor 28a/(AT)\rfloor + 1$ and $M = \lfloor 172032A^4H^6/a^3\rfloor$. If (11) holds, then*

$$T \le 16\,(2M+1)^7H^2.$$

Proof. Call a start i *good* if $s \le i$, $i+9 \le s+T$ and $k_i,\dots,k_{i+6} \le H$.

Most starts are good. Let $\mathcal{B}$ be the set of $m \in [s,s+T-1)$ with $k_m > H$. Then $|\mathcal{B}|(H+1) \le \sum_m k_m = q_{s+T-1} - q_s \le a/A$. Since $H \cdot AT > 28a$, this gives $28|\mathcal{B}| \le T$. Each $m \in \mathcal{B}$ spoils at most seven starts, so at least $T - 9 - 7|\mathcal{B}| \ge T/2$ starts are good, as $T \ge 36$.

Good starts satisfy Proposition 11.7. Let i be good. Then $i+8 < N$, and $b_i, b_{i+1}, b_{i+2} < 2A$. So Lemma 11.5 applies at i, $i+1$ and $i+2$. Lemma 11.6 at i and at $i+1$ bounds all seven entries of $\theta(i)$ by M. Therefore $T/2 \le 8(2M+1)^7H^2$. □

11.7. **Proof of Theorem 1.4.**

Proposition 11.9. *Let $T \geq 36$ be the length of a window run in $[A, 2A)$. Then $T^{77} \leq C_{77}\Delta^{22}a^{23}$, where*

$$C_{77} = 122449^{20} + (8 \cdot 237568 \cdot 56^5)^{11} + (64 \cdot 50176)^{11} + 8^{16} \cdot 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}.$$

Proof. We use Proposition 7.3, $T^6A^2 \leq 96 \cdot 10^6 a^3$, and Proposition 5.1, which gives $T^2 \leq 2T(T-1) \leq 8A\Delta$.

Case $AT > 28a$. Then $784a^2T^4 \leq T^6A^2 \leq 96 \cdot 10^6 a^3$, so $T^4 \leq 122449\,a$ and $T^{77} \leq 122449^{20}a^{20}$.

From now on $AT \leq 28a$, so $H \cdot AT \leq 28a + AT \leq 56a$.

Case (11) *fails.* Then $a^3(AT)^5 < 237568A^4(H\,AT)^5 \leq 237568 \cdot 56^5 A^4a^5$, so $AT^5 \leq 237568 \cdot 56^5 a^2$. Hence $T^7 = T^5T^2 \leq 8\Delta\,AT^5 \leq 8 \cdot 237568 \cdot 56^5\,\Delta a^2$. Raise this to the eleventh power.

Case (11) *holds and $M = 0$.* Then $T \leq 16H^2$, so $T(AT)^2 \leq 16(56a)^2$. Hence $T^7 = T^3(T^2)^2 \leq 64\Delta^2\,T^3A^2 \leq 64 \cdot 50176\,\Delta^2a^2$.

Case (11) *holds and $M \geq 1$.* Then $2M + 1 \leq 3M$ and $Ma^3 \leq 172032A^4H^6$, so

$$T\,a^{21} \leq 16 \cdot 3^7\,(Ma^3)^7H^2 \leq 16 \cdot 3^7 \cdot 172032^7\,A^{28}H^{44}.$$

Multiplying by $(AT)^{44}$ and using $H \cdot AT \leq 56a$ gives $T^{45}A^{16} \leq 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}\,a^{23}$. Finally

$$T^{77} = T^{45}(T^2)^{16} \leq 8^{16}\Delta^{16}\,T^{45}A^{16} \leq 8^{16} \cdot 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}\,\Delta^{16}a^{23}.$$

In every case $T^{77} \leq C_{77}\Delta^{22}a^{23}$. □

Proof of Theorem 1.4. This is the argument of Theorem 1.1, with Proposition 11.9 in place of Corollary 5.3. Let $\varepsilon > 0$ and choose an integer $m > 1/\varepsilon$. By Lemma 2.2, $\Delta^{22m} \leq C_{22m} \cdot 32a^7$, so every dyadic window has $T \leq 35$ or

$$T^{77m} \leq C_{77}^m\,C_{22m}\,32\;a^{23m+7}.$$

Since $(23m + 7)/(77m) = 23/77 + 1/(11m) \leq 23/77 + \varepsilon/2$, each of the $\lfloor \log_2 a \rfloor + 1$ windows has $T \ll_\varepsilon a^{23/77+\varepsilon/2}$, and the theorem follows. □

Remark 11.10. In terms of the scales, Proposition 11.8 says $T \ll X^7H^2$, where $X = A^4H^6/a^3$ bounds the seven pattern integers and $H \asymp a/(AT)$. Thus $X \asymp a^3/(A^2T^6)$ and $T^{45}A^{16} \ll a^{23}$. The drop bound $A \gg T^2/\Delta$ then gives $T^{77} \ll \Delta^{16}a^{23}$. The scale condition (11) says $Q^4 \gg aH^5$, where $Q = a/A$, and its failure gives the stronger bound $T \ll a^{2/7+o(1)}$.

12. The Lean formalisation

The repository accompanying this paper contains a Lean 4 project (toolchain `v4.35.0`, Mathlib `v4.35.0`) that proves Theorem 1.1. The final statement is

```
theorem P_le : ∀ ε > (0 : ℝ), ∃ C > 0, ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ ((3 : ℝ) / 10 + ε)
```

and Theorem 1.4 is

```
theorem P_le_23_77 : ∀ ε > (0 : ℝ), ∃ C > 0, ∀ a b : ℕ, 0 < b → b <
a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ ((23 : ℝ) / 77 + ε)
```

Here `P a b` is defined by well-founded recursion as the number of positive terms ($P(a, 0) = 0$, $P(a, b) = P(a, a \bmod b) + 1$ for $b > 0$), and the file checks $P(35, 22) = 7$. Theorem 1.2 is formalised as

```
theorem P_le_loglog (hNT : NairTenenbaumDelta)
    (hBT : BretecheTenenbaumMean) :
    ∃ C > 0, ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ (3 / 10 : ℝ)
        * √(Real.log (Real.log (3 * a)))
```

Theorems 1.3 and 10.2 are formalised as

```
theorem P_le_rises : ∃ C > 0, ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ (1 / 4 : ℝ) * √(rises a b + 1)
theorem hankelInv_pos_int (hba : b < a) (hi : i < P a b)
    (hr : r ≤ P a b - i) :
    ∃ n : ℕ, 0 < n ∧ hankelInv a b i r = n
```

together with `hankelInv_one`, …, `hankelInv_four`; the last states `hankelInv a b i 4 = sixL` …, the six-step integer of Lemma 7.1. Here `NairTenenbaumDelta` and `BretecheTenenbaumMean` are Lean propositions stating (9) (with $\varepsilon = 1/4$) and (10), with existential constants and thresholds. `hooleyDelta` is defined as the supremum over $u \in \mathbb{R}$, exactly as above. For `P_le`, `P_le_23_77`, `six_step_bound`, `P_le_of_shortDivisorBound`, `P_le_loglog`, `P_le_rises`, `hankelInv_pos_int` and `hankelInv_four`, the command `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`.

Paper	Lean	File
Definition of $P(a,b)$	`P`, `orb`	`Defs.lean`
Lemma 2.1	`div_lt_div_mod`	`Block.lean`
Lemma 3.2	`curvature_identity`, `one_le_bD`	`Block.lean`
Lemma 3.3	`fiber_identity`, `block_fiber_eq`	`Block.lean`
Lemma 3.4	`bu_mul_le`, `bv_mul_le`, `bD_mul_le`	`Block.lean`
Lemma 4.1	`fiber_poly_identity`, `fiber_card_le`	`Fiber.lean`
Window runs	`Window`, `DivBound`	`Window.lean`
Proposition 5.1	`drop_bound`	`Window.lean`
Proposition 5.2	`curvature_bound`	`Window.lean`
Corollary 5.3	`window_bound`	`Window.lean`
Dyadic decomposition	`P_eq_sum_visits`, `visits_window`	`Main.lean`
Theorem 1.1	`P_le`	`Main.lean`
Lemma 2.2	`exists_divisors_power_bound`	`DivisorPowerBound.lean`
Lemma 7.1	`six_divisibility`, `six_identity`	`SixStep.lean`
Corollary 7.2	`six_core`	`SixStep.lean`
Proposition 7.3	`six_step_bound`	`SixStep.lean`
Proposition 8.1	`drop_count_bound'`	`ShortSum.lean`
Geometric sum	`sum_min_le`	`Geometric.lean`
Theorem 8.3	`P_le_of_shortDivisorBound`	`Conditional.lean`
Δ_{H}, $\tau_A \le \Delta_{\mathrm{H}}$	`hooleyDelta`, `dcount_le_hooleyDelta`	`Hooley.lean`
(9), (10)	`NairTenenbaumDelta`, `BretecheTenenbaumMean`	`Hooley.lean`
Lemma 8.4	`sum_hooleyDelta_div_le`	`Hooley.lean`
Theorem 1.2	`P_le_loglog`	`Hooley.lean`
Lemma 9.1	`ap_identity`, `ap_bound`	`Rises.lean`
Lemma 9.3	`drop_lt_of_not_rise`	`Rises.lean`
Lemma 9.4	`sq_card_le_of_fiber`	`Rises.lean`
Proposition 9.5	`rises_drop_bound`, `rises_quot_bound`, `rises_incr_bound`	`Rises.lean`
General geometric sum	`sum_min_le_gen`	`Geometric.lean`
Theorem 1.3	`P_le_rises`, `P_le_of_no_rises`	`RisesMain.lean`
Lemma 10.1	`wt_spec`, `hm_zero`, `hm_succ`, `hm_int`	`Hankel.lean`
Theorem 10.2	`hankelInv_pos_int`, `hankelInv_one`, …, `hankelInv_four`	`Hankel.lean`
Hadamard's inequality	`det_le_prod_diag`	`Hadamard.lean`
μ_i, ν_i, shift rule	`muF`, `nuF`, `muF_nb_mul`	`Functional.lean`
Lemma 11.5	`hankelInv_le_prod`, `label_bound`	`Gram.lean`
Lemma 11.1	`null_k`, `null_r`	`TwoState.lean`
Lemma 11.3	`quad_root`	`TwoState.lean`
Lemma 11.4	`mult_identity`, `quart_eq`	`TwoState.lean`
Lemma 11.6	`gramP_le`, `state_bound`	`PatternBound.lean`
Proposition 11.7	`pattern_count`	`PatternCount.lean`
Proposition 11.8	`good_card`, `two_state_window`	`TwoStateMain.lean`
Proposition 11.9	`window_77`, `visits_77`	`TwoStateMain.lean`
Theorem 1.4	`P_le_23_77`	`TwoStateMain.lean`

Some details of the formalisation:

- The algebraic identities (Lemmas 3.2, 3.3 and (7)) are proved over $\mathbb{Z}$ by the tactic `linear_combination`, using the explicit coefficients above. They are then transferred to $\mathbb{N}$ by writing $z = w + f$, $y = z + e$, $x = y + d$, which avoids truncated subtraction.

- The curvature is the natural number `bD a x` $= (x-y)v-(y-z)u$. Lean first proves $eu < dv$, so this truncated subtraction is the true value.
- The counting arguments use `Finset.card_le_mul_card_image_of_maps_to`: the number of good indices is at most the size of the parameter box times the largest fiber.
- The divisor bound $\tau(n)^k \le C_k n$ (Lemma 2.2) is the file `DivisorPowerBound.lean` from the `openai/math` repository [8] (Apache-2.0), vendored unchanged except for its import lines.
- Lemma 7.1(2) is checked by `linear_combination` with the coefficients given in its proof, which were found by polynomial division. Corollary 7.2 proves $L \ge 1$ directly from the identity, since L is an integer by definition.
- The identities $I_2 = d$, $I_3 = D$, $I_4 = L$ are checked by `linear_combination`, with certificates generated by polynomial division. The moment entries $h_i(s)$ are computed in Lean from the recursion of Lemma 10.1(3).
- Mathlib has no Hadamard inequality for determinants, so `Hadamard.lean` proves it: rescale to unit diagonal, and use $\prod \lambda_j \le \prod e^{\lambda_j - 1} = e^{\mathrm{tr}\, - n} = 1$ for the eigenvalues.
- The identities of Lemmas 11.1, 11.3 and 11.4 are checked by the tactics `ring` and `linear_combination`. The certificate for Lemma 11.4 has about 350 terms and was found by polynomial division. Root counting uses the Mathlib lemma `card_roots'`.
- The project has about 5800 lines of Lean. `lake build` rebuilds it in a few minutes once Mathlib is built.

13. Limits of the method

Proposition 7.3 removes the divisor loss from the local bound, but on its own the exponent is still $3/10$. The local bound $T \ll a^{1/2}A^{-1/3}$ and the drop bound $T \ll A^{1/2}$ (up to the divisor factor) both allow $T \asymp a^{3/10}$ at $A \asymp a^{3/5}$. Even an ideal short-interval divisor estimate with G constant leaves this balance unchanged. So sharpening constants or improving the divisor counts cannot give a logarithmic bound. Further progress needs to restrict *which* configurations of the parameters (u, v, w, z, D, E, F) can occur together along a single orbit. Theorem 1.3 is one restriction of this kind: a long orbit must have many rises in its quotient increments. Theorem 10.2 supplies the natural candidates for further ones, the higher invariants I_r. Section 11 uses I_4 in exactly this way.

The two-state count of Section 11 loses only through the number X^7 of patterns. With patterns at four consecutive states one has thirteen integers and as many equations as unknown increments. If each such pattern allowed only $O(1)$ configurations, the same optimisation would give the exponent $39/131$. But the four-state equations have positive-dimensional families of solutions. For instance, the Newton coefficients $(\alpha, -3\alpha-1, 3\alpha+3, -\alpha-3, 1)$, with coefficient polynomial $(z-1)^3(z-\alpha)$, at every state, together with constant increments $k_j = t$, satisfy all the linear and quadratic relations for every t. Such families are excluded only by the remainder equations, which the count above discards. A refinement through the degree-five orthogonal polynomial and the curve bounds of [2] has been proposed to reach the exponent $14/47$. We have not checked its case analysis and do not include it.

There is no obvious monotonicity to exploit. The curvatures D_i of successive blocks need not decrease. For $a = 13242$ the orbit of 255 begins

$$255,\ 237,\ 207,\ 201,\ 177,\ 144,\ 138,\ 132,\ \ldots$$

with quotients $51, 55, 63, 65, 74, 91, 95, 100, \ldots$. The curvatures of the first five blocks are $24, 12, 6, 111, 30$, all for blocks that lie within a factor of two.

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